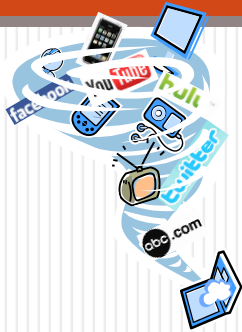


CA 7101 User Interface Design



Chutisant Kerdvibulvech, Ph.D.
Associate Professor

รศ.ดร.ชุตินันต์ เกติวิบูลย์เวช

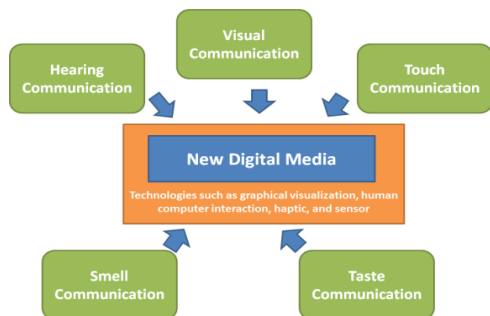
L2

1

New Digital Media

2

New Digital Media



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Visual Communication

Augmented Reality



Sixth Sense (MIT)

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Augmented Reality Magic (card trick), Newmagic Communications Inc.

TED
TALKS

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Touch Communication



A blue fancy robotic bear was designed to relieve pain for kids in the Wired Magazine [Fal et al. 2015]

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Smell Communication

A smell-based system called Dollars & Scents inside the entrance to their laboratory at MIT

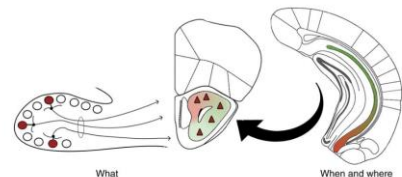
- The system releases scents and perfumes into the air differently depending on stock market changes.
- Emit a lemon-like fragrance if NASDAQ index is going down, while they emit a rose-like fragrance if the stock market is going up.
- Similarly, they also built an olfaction system, called inStink, for conveying ambient presence and activity awareness for food.

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Smell Communication

- Scientists uncover new connection between smell and memory
- Findings offer opportunities for improved smell tests in Alzheimer's disease diagnosis



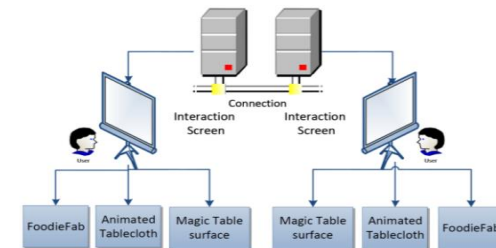
Conceptual diagram for the formation of episodic odour memory. A model of episodic odour memory whereby information regarding odour quality and spatiotemporal context merge at the level of the AON producing cellular populations that represent previously encountered odours (what) within the context in which they occurred (when and where)

Alif J. Aqrabawi, Jun Chul Kim. **Hippocampal projections to the anterior olfactory nucleus differentially convey spatiotemporal information during episodic odour memory.** *Nature Communications*, 2018; 9 (1)

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Taste Communication

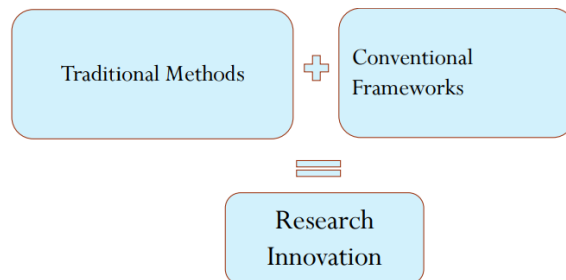


The Food Media's system configuration is set up by [Wei 2013] from National University of Singapore.

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Integration leads to Innovation



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Integration AR with Music

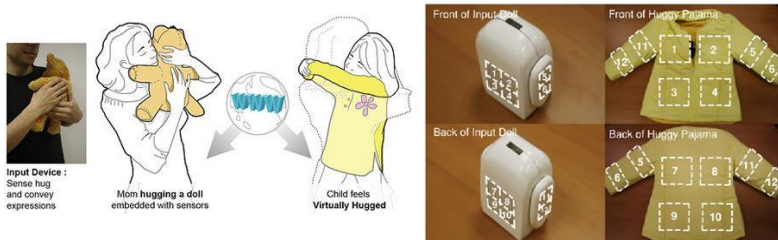
- Integration AR with Music for entertaining and helping people



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From Huggy Pajamas to T-Jacket

- Touch Communication: Application



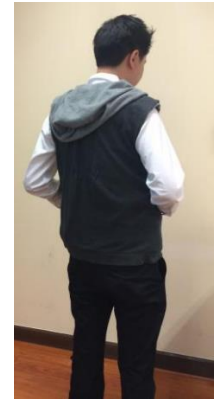
Smart Clothes as a Tangible User Interface to Affect Human Emotions using Haptic Actuators, F. Arafsha (2012)

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T-Jacket

- Wearing T-Jacket for Touch communication
 - Controlled by my smartphone (iPhone) using Bluetooth
 - Charged by a normal voltage
 - Mobilities
 - Children



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Integrating T-Jacket with Visual

- The integrated system between touch and visual communications using a built cardboard headset



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Griffin, Ben; Baston, Laurel. "Interfaces" (Presentation): 5. Archived from the original on 14 July 2014. Retrieved 7 June 2014. The user interface of a mechanical system, a vehicle or an industrial installation is sometimes referred to as the human-machine interface (HMI)

What is User Interface (UI)?

- The user interface (UI), in the industrial design field of human-computer interaction, is the space where interactions between humans and machines occur.



Griffin, Ben; Baston, Laurel. "Interfaces" (Presentation): 5. Archived from the original on 14 July 2014. Retrieved 7 June 2014. The user interface of a mechanical system, a vehicle or an industrial installation is sometimes referred to as the human-machine interface (HMI)

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What is User Experience (UX) Design?

- The process design teams use to create products that provide meaningful and relevant experiences to users

<https://www.interaction-design.org/literature/topics/ux-design>

- Including the practical, experiential, affective, meaningful and valuable aspects of human–computer interaction and product ownership



UI

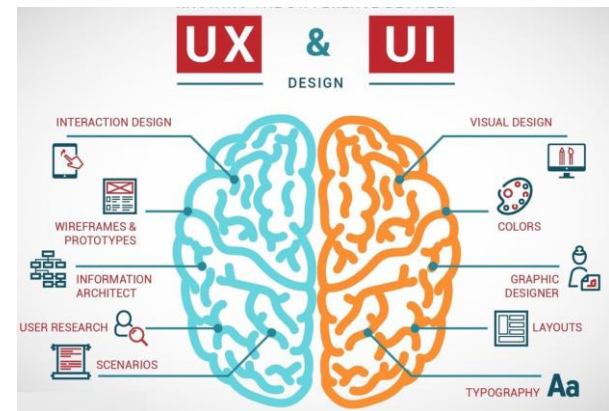
UX

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<https://medium.com/@jgreenhalgh>

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UX and UI

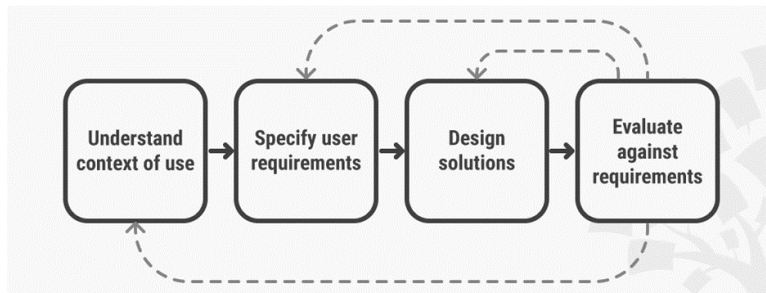


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User-centered design

- An iterative process where you take an understanding of the users and their context as a starting point for all design and development <https://www.interaction-design.org/literature/topics/ux-design>



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Conclusion

- New Digital Media
- Five Senses Communication by Integration
- Touch Technology

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Questions?

Assoc. Prof. Dr. Chutisant Kerdvibulvech
Graduate School of Communication Arts and Management Innovation,
National Institute of Development Administration (NIDA)
chutisant.ker@nida.ac.th
<https://www.facebook.com/chutisant.kerdvibulvech>

Assoc. Prof. Dr. Chutisant Kerdvibulvech